Chapter 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science concerned with creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, understanding language, recognizing patterns, and making decisions. AI systems can be rule-based or can learn from data. Machine learning is one of the most important approaches used in modern AI.

AI applications include recommendation systems, virtual assistants, search engines, fraud detection, image recognition, and educational software. A typical AI system receives input, processes information, and produces an output or decision. The quality of an AI system depends on factors such as data quality, algorithms, computing resources, and evaluation methods.